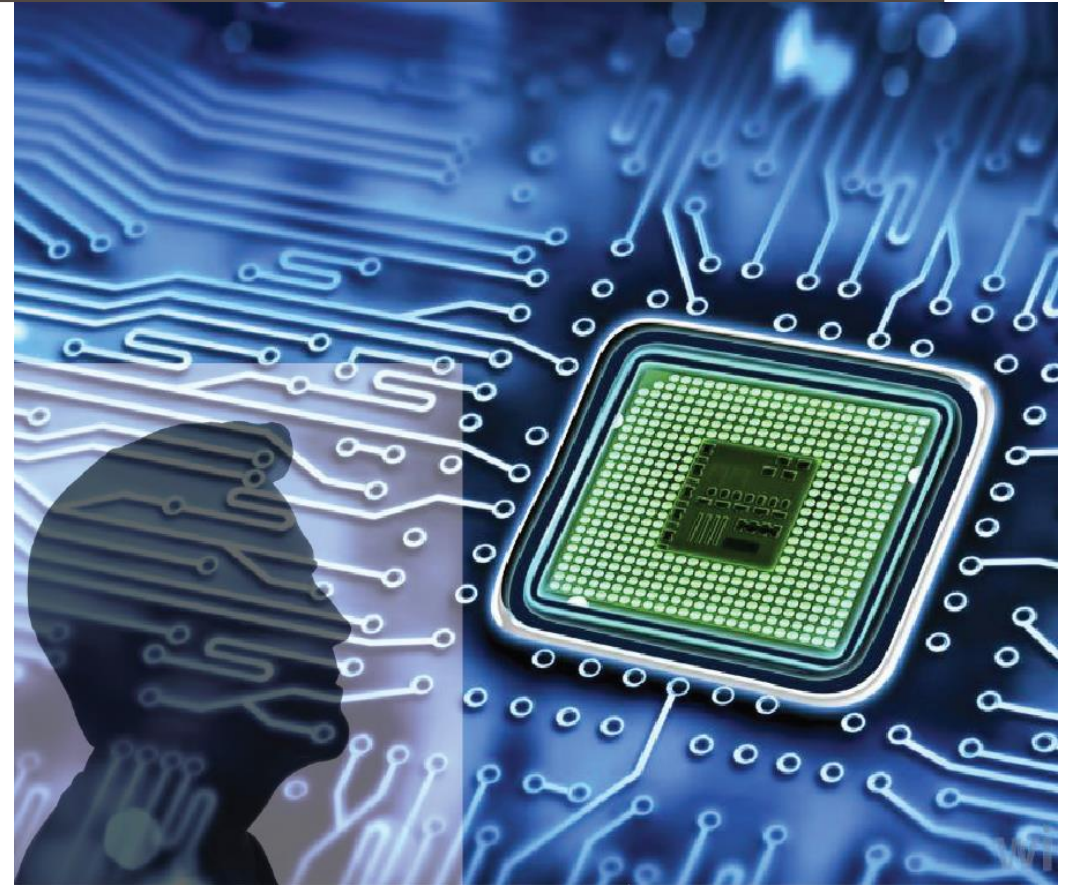


VLSI DESIGN FLOW

Week-12 Lecture-2 Physical Design

Sneh Saurabh
13th November, 2024

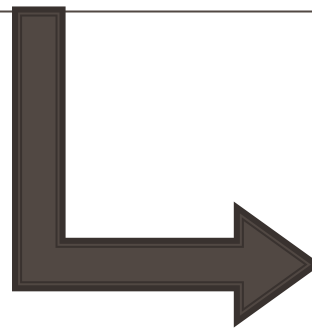


VDF: Revision/Announcements

-

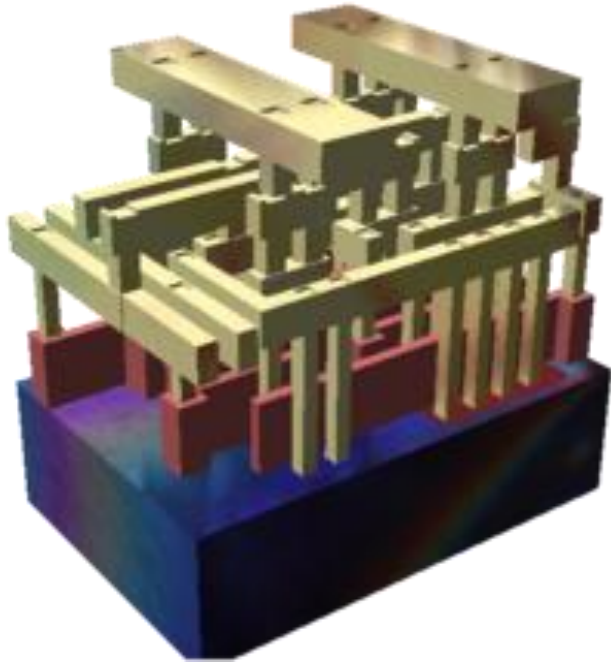


Physical Concepts for Physical Design



Interconnects/Wires

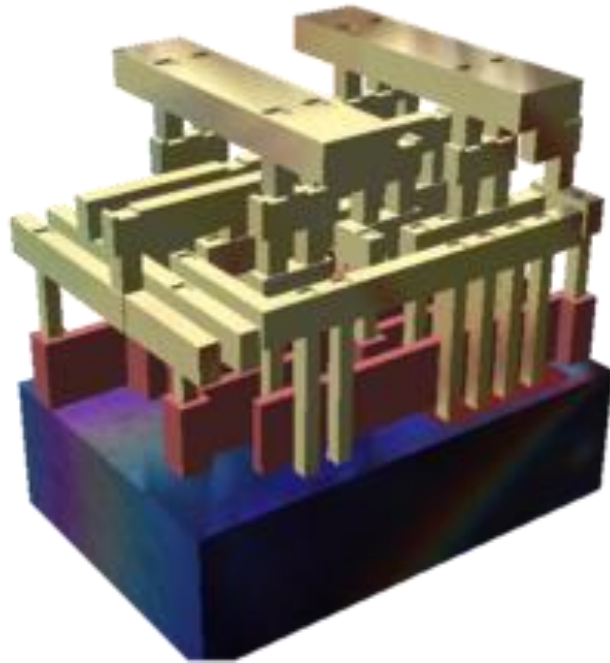
Nature of Interconnects/Wires in Integrated Circuits



Source: Wikipedia

- Interconnects fabricated as layers of metals
- Properties of different layers of metals can be different
- Height of a metal layer is fixed and expected to be uniform (except due to unavoidable process-induced variations)

Nature of Interconnects/Wires in Integrated Circuits



Source: Wikipedia

Metal Layers Two Types:

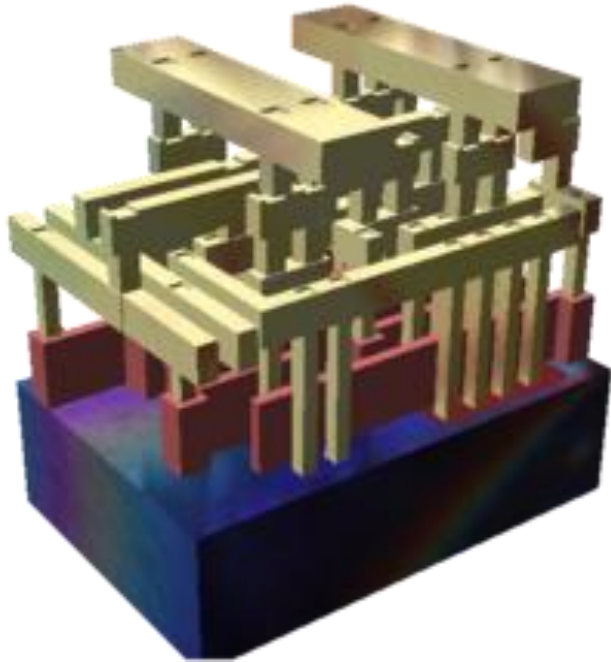
➤ **Wiring layers**

- Metal layers making interconnection
- Parallel to wafer surface

➤ **Via Layers**

- Connects wiring layers
- Perpendicular to wafer surface
- Consist of square shapes that connect overlapping wiring segments of neighboring wiring layers

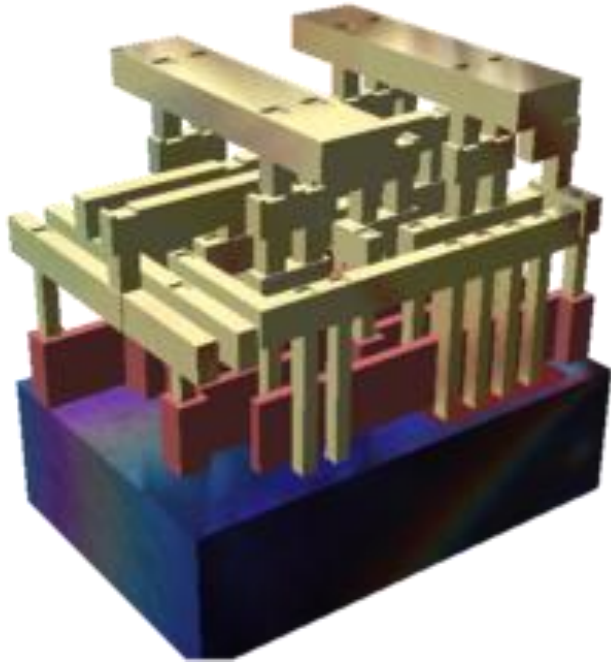
Nature of Interconnects/Wires in Integrated Circuits



Source: Wikipedia

- Shape of wires in the layout decide where metal needs to be deposited during fabrication
- Interconnect shapes are **3D-polyhedral**
- Interconnects have uniform thickness (height)
 - Can be represented as 2D polygons

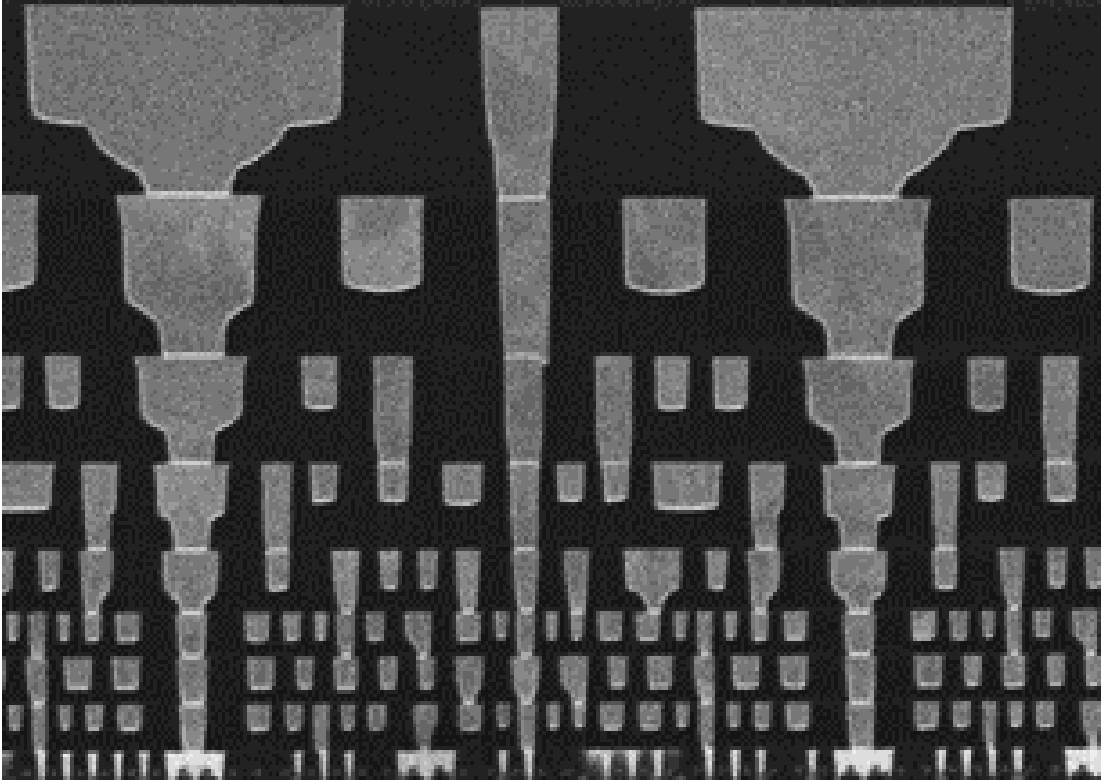
Wiring Layers: Direction



Source: Wikipedia

- Each Wiring Layer has a **preferred direction**
 - Vertical or Horizontal
- Each successive wiring layers alternate between Vertical and Horizontal direction
 - Can lead to increase in via count

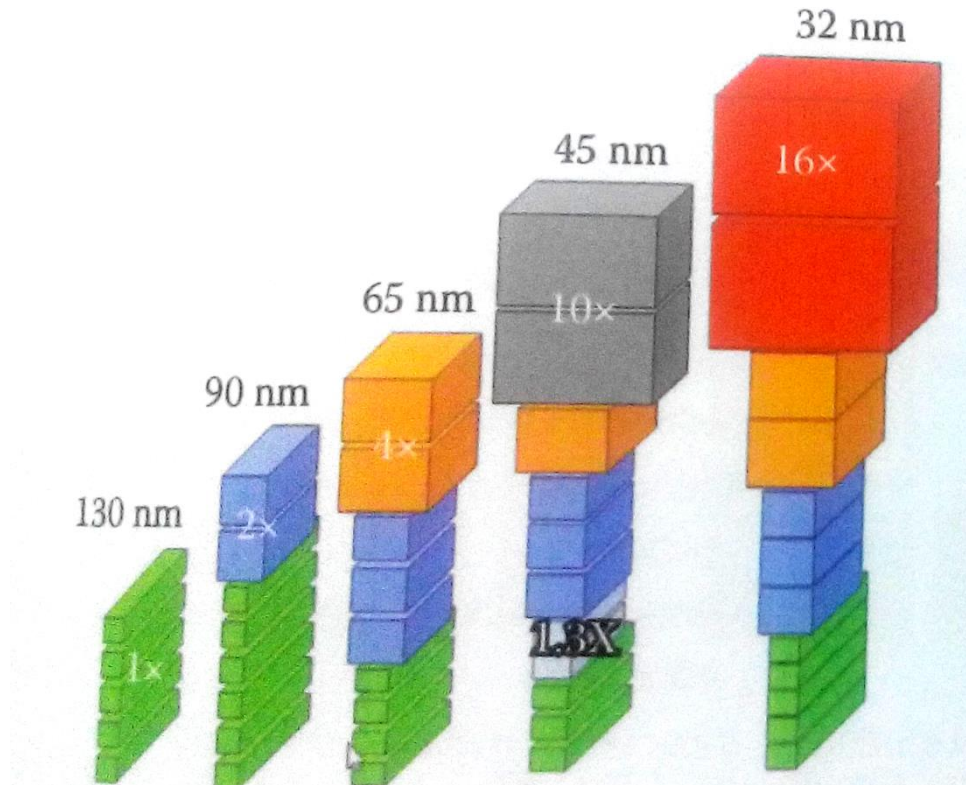
Wiring Layers: Height



- Height of wiring layers can be different in different layers
- In general: metal height **increases from the bottom to the top** of an IC

Source: “Process and Electrical Results for the On-die Interconnect Stack for Intel’s 45nm Process Generation,” *Intel Technology Journal*, Volume 12, Issue 2, 2008

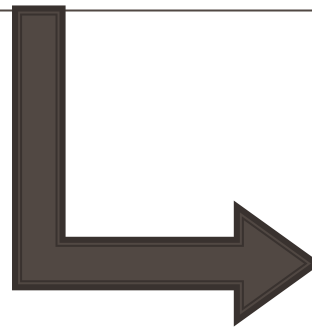
Wiring Layers: Height



Source: Electronic Design Automation for Integrated Circuits Handbook, Second Edition, Luciano Lavagno, Igor L. Markov, Grant E. Martin, Louis K. Scheffer, - 2016, - CRC Press

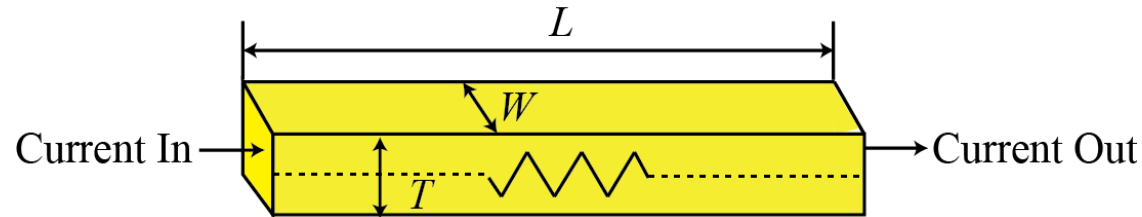
- High performance IC can use more than 15 metal layers
- Metal Layers are separated by dielectrics : generally SiO_2
- Low-k material can also be used
 - Example: Carbon-doped oxide low-k dielectric
- Reliability an important concern for metal layers

Basic Concepts for Physical Design



Interconnect
Parasitics

Interconnect: Resistance



- Resistance:

$$R = \rho \frac{L}{TW}$$

- $R = R_s \frac{L}{W}$ where $R_s = \frac{\rho}{T}$ is called the sheet resistance
- Sheet resistance is defined in the library

- At high frequency, the resistance of an interconnect tends to increase
 - Skin effect: current tends to flow primarily on the conductor's surface
 - Important for wider and thicker wires at the top metal layers (clock lines that work at high frequency)

Interconnect: Capacitance

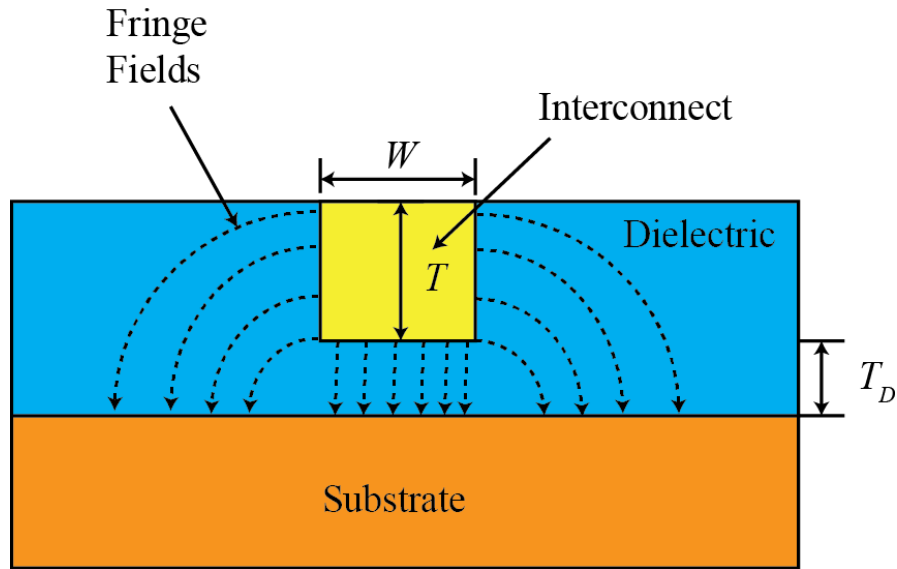
Origin of interconnect Capacitance:

- Interconnects lie within a dielectric material
- Electric potential of interconnects changes during circuit operation
 - Electric field and the stored charges in the surrounding dielectric material change.
 - Consequently, an interconnect exhibits substantial capacitance

Factors influencing Interconnect Capacitance:

- Depends on their geometry, environment (the location and geometry of other interconnects), and the property of the surrounding dielectric.
- The computation of interconnect capacitance is a nontrivial problem.

Interconnect: Capacitance of Strip

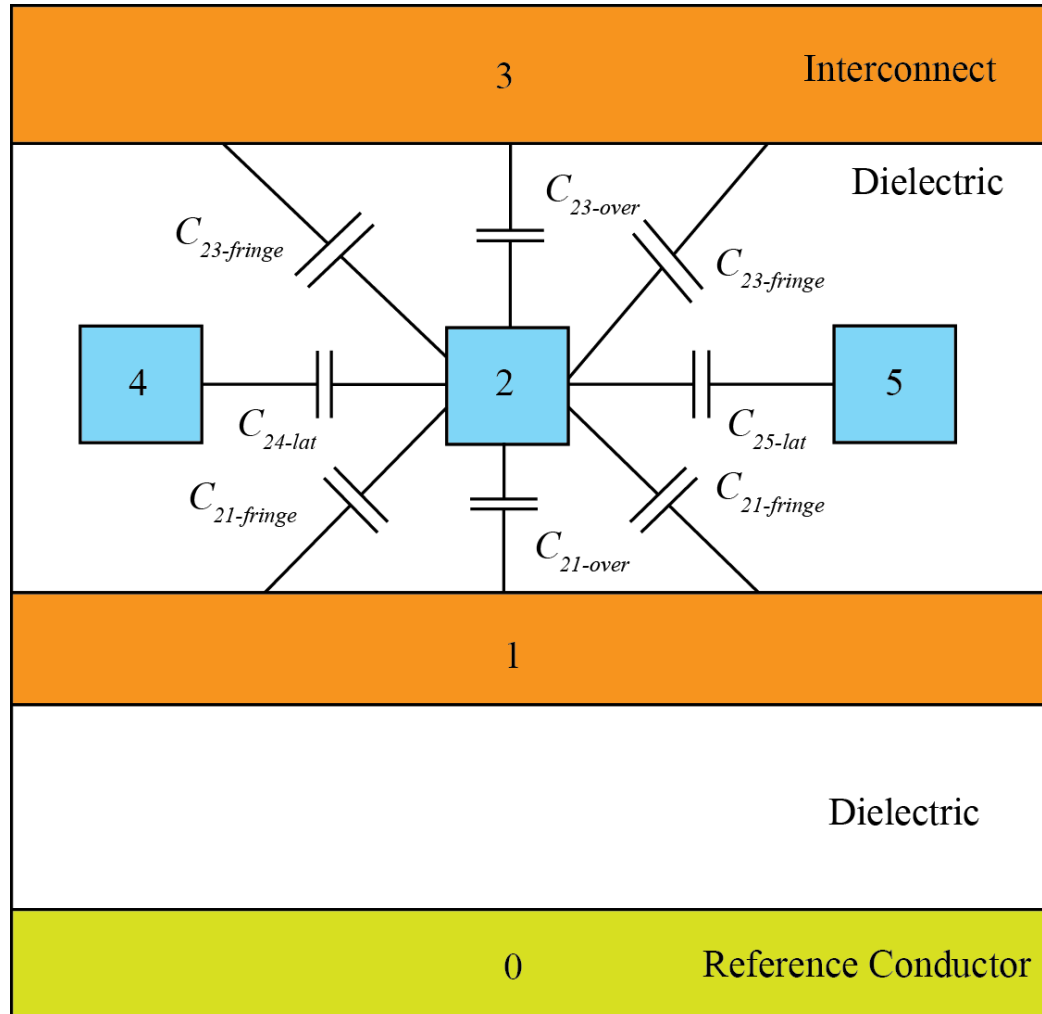


Two Components:

1. Parallel plate component: $C_{pp} = \frac{\epsilon_d \epsilon_0}{T_D} WL$
2. Fringe component: field lines that emanate from the sidewalls

- Both these components can be added together
- With technology advancement, fringe capacitance is increasing

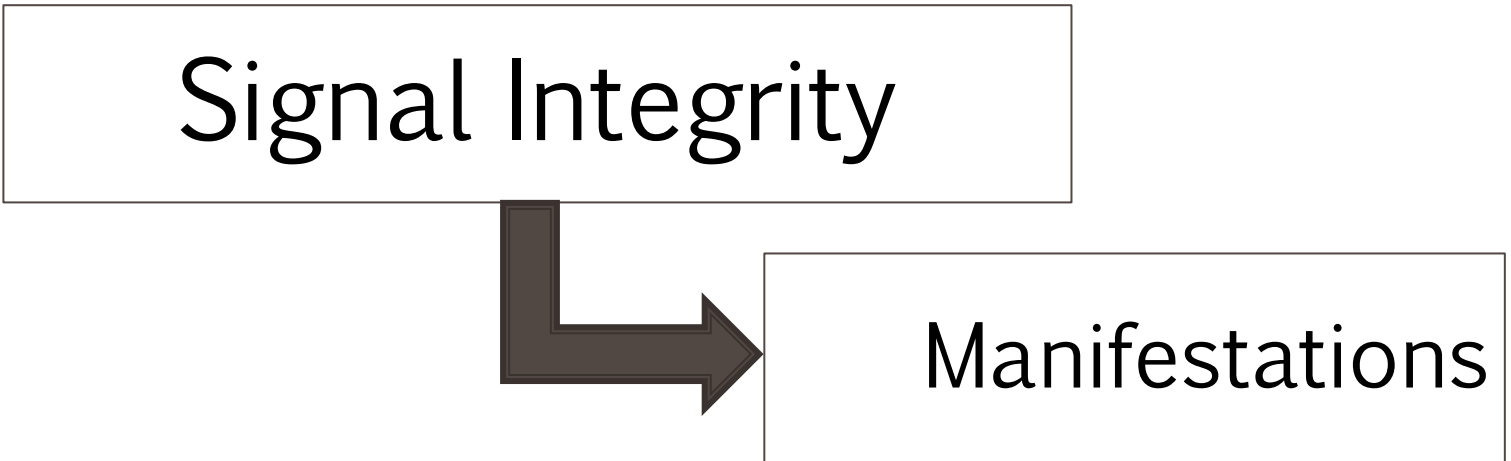
Multiple Interconnects: Capacitance



- Electric field lines emanating from an interconnect are modified by the neighboring interconnects in a complicated manner

Three major components of capacitance:

1. **Overlap capacitance:** due to the overlap between two conductors in different planes.
2. **Lateral capacitance:** formed by two parallel edges of nonoverlapping conductors in the same plane.
3. **Fringe capacitance:** between two conductors in different planes due to electric fields originating from the sidewalls



```
graph TD; A[Signal Integrity] --> B[Manifestations]
```

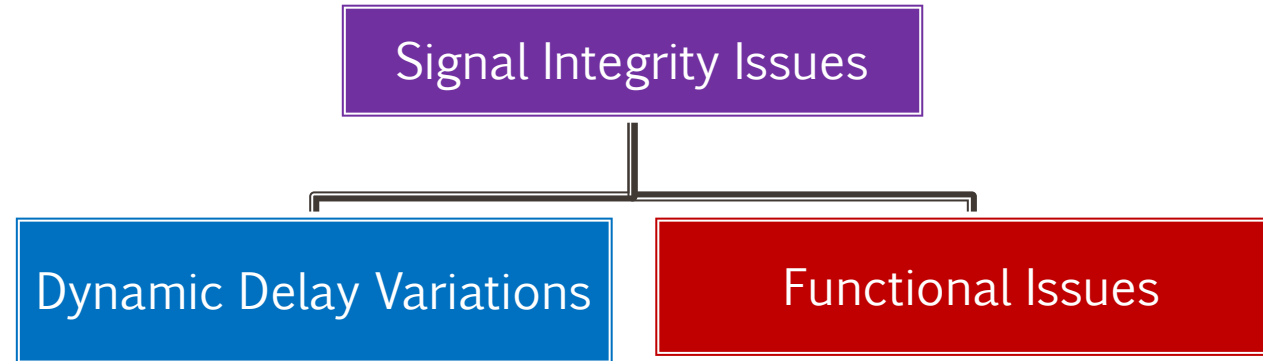
Signal Integrity

Manifestations

Effects of Coupling of Interconnects

Impact of coupling capacitance among interconnects

- Voltage in one interconnect can impact the voltage in the other
- Can create signal integrity problems in a circuit



Signal Integrity



Dynamic Delay
Variations

What is base delay?

Consider the portion of a circuit

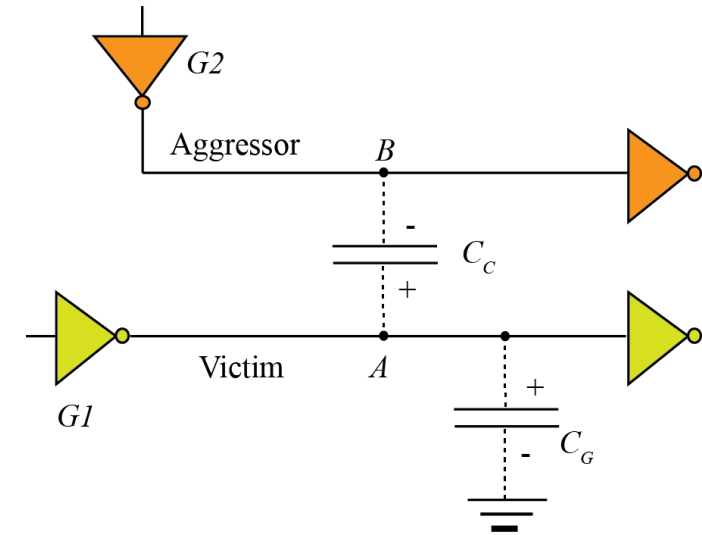
- Let us assume A is the victim
 - The signal line at which we observe the signal integrity issue is known as the victim line
- Assume gate G1 (driver of the victim net A) makes a 0→1 transition at the output.
 - The driver G1 provides charge for the ground capacitor C_G to change from 0 to V_{DD} .

Case1: B is held constant to 0

- G1 needs to provide a charge for the coupling capacitance C_c to change from 0 to V_{DD} .
- Hence, G1 needs to charge $C_G + C_c$.

Case2: B is held constant to 1

- Initially charge on C_c is $-V_{DD}$ and finally 0
- Hence, G1 needs to charge $C_G + C_c$.



- The delay computed by assuming aggressors are held to constant logic is known as **base delay**.
- The base delay is same whether aggressor is at 0 or 1
- Base delay is obtained by ignoring SI effects

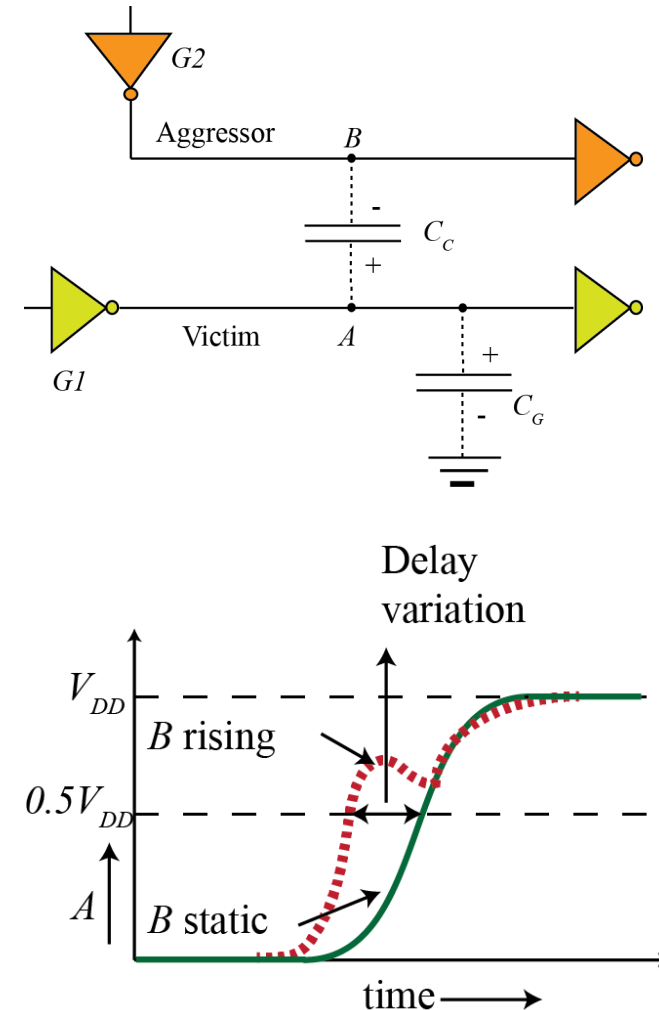
How is dynamic delay variations created?

- Assume gate G1 (driver of the victim net A) makes a 0→1 transition at the output.

Case1: B also makes a 0→1 transition while A was transitioning:

- Voltage across C_c is 0 V, both initially and finally.
- Hence, less charge needs to be provided by G1 for charging C_c
- Delay of G1 will decrease.

- This effect is modeled as *negative incremental delay* over base delay



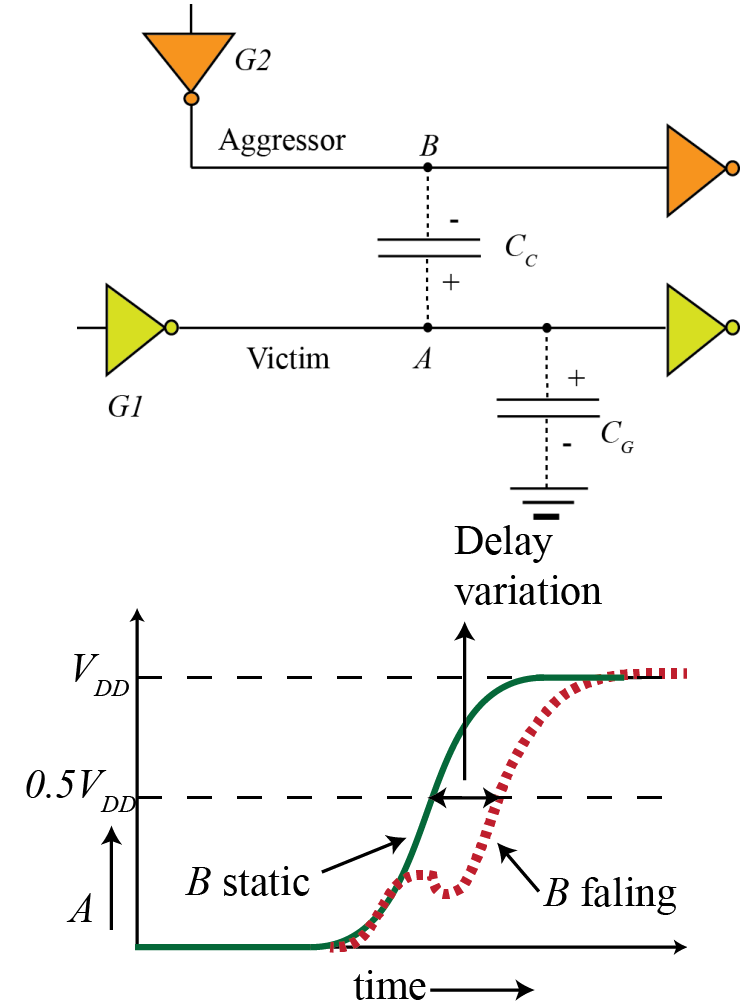
How is dynamic delay variations created?

- Assume gate G1 (driver of the victim net A) makes a 0→1 transition at the output.

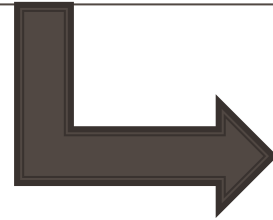
Case2: B makes a 1→0 transition while A was transitioning:

- Voltage across C_c is $-V_{DD}$ initially and $+V_{DD}$ finally.
- Hence, in effect, $2C_c$ needs to be charged by G1
- Delay of G1 will increase.

- This effect is modeled as *positive incremental delay* over base delay



Signal Integrity



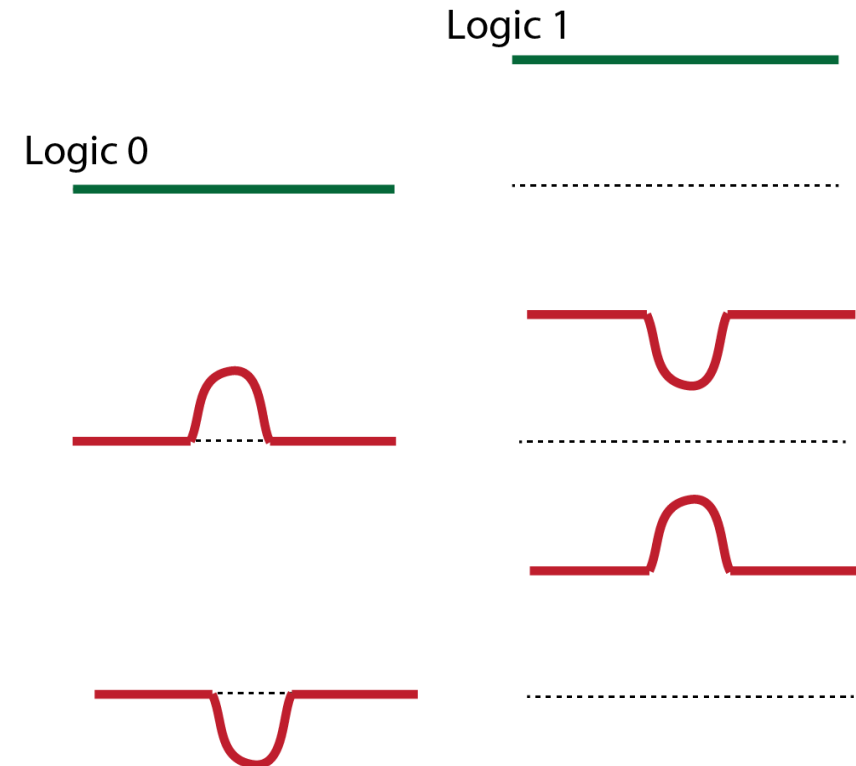
Noise or Glitches

What is a Glitch?

- A glitch can occur in a victim net held at constant logic value due to a transition in the aggressor net.
- Glitches are also known as crosstalk noises

Depending on the value of the victim net and the type of transition at the aggressor nets, four types of glitches are possible:

- a. Logic 0 has a rise glitch,
- b. Logic 0 has an undershoot
- c. Logic 1 has a fall glitch
- d. Logic 1 has an overshoot.



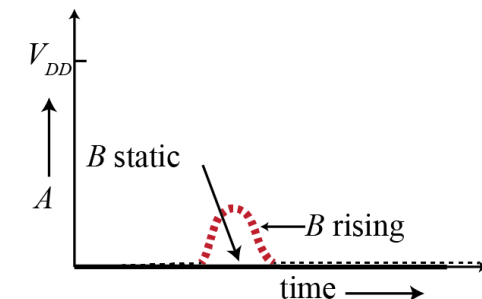
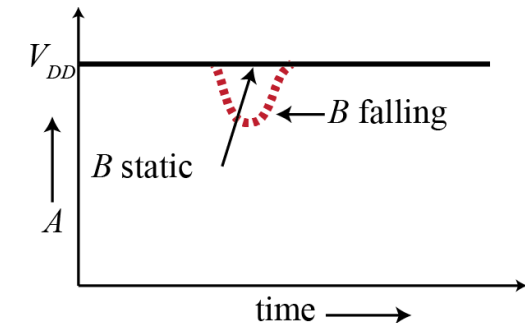
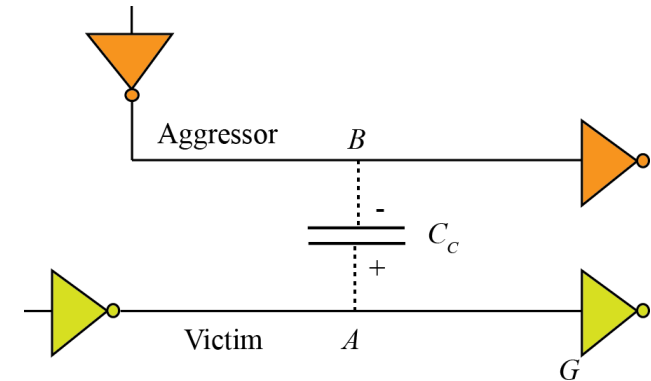
How are Glitches Created?

Consider the portion of a circuit

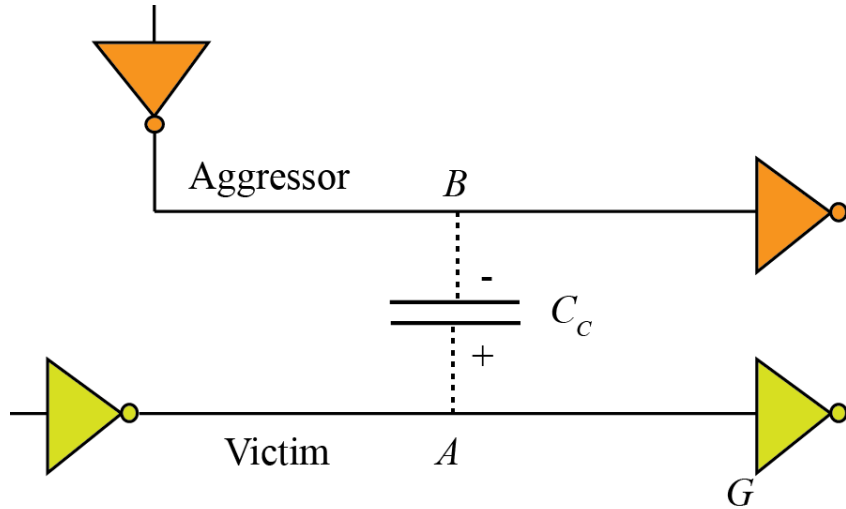
- Let us assume A is the victim
- Assume that A is at a steady state of V_{DD} [logic 1]
- Assume that line B is also at the steady state of V_{DD} [logic 1]
- Voltage across the coupling capacitor C_C is 0 V.

- Assume that the neighboring line B makes a quick transition from V_{DD} to 0 V
- The coupling capacitor C_C now gets charged to $+V_{DD}$ due to voltage difference in lines A and B
- The driver of the line A will supply this charge, and it will cause a temporary dip in the voltage of the line A

- Similarly, when the aggressor (line B) makes a 0 to V_{DD} transition, there can be a bump in the voltage of the victim (line A) that was held at logic 0



On what factors the magnitude of glitches depend?



Slew of the aggressor net

- Glitch increases when the aggressor transitions quickly.

Coupling capacitance between the aggressor and the victim net:

- Glitch increases with the increase in the coupling capacitance.

Ground capacitance of the victim net

- Glitch decreases as the ground capacitance of the victim increases.

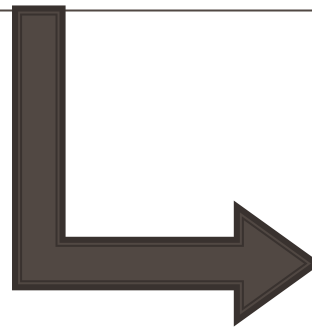
Strength of the driver of the victim net

- Glitch decreases as the strength of the driver of the victim net increases.

When does Glitches Cause Functional Problems?

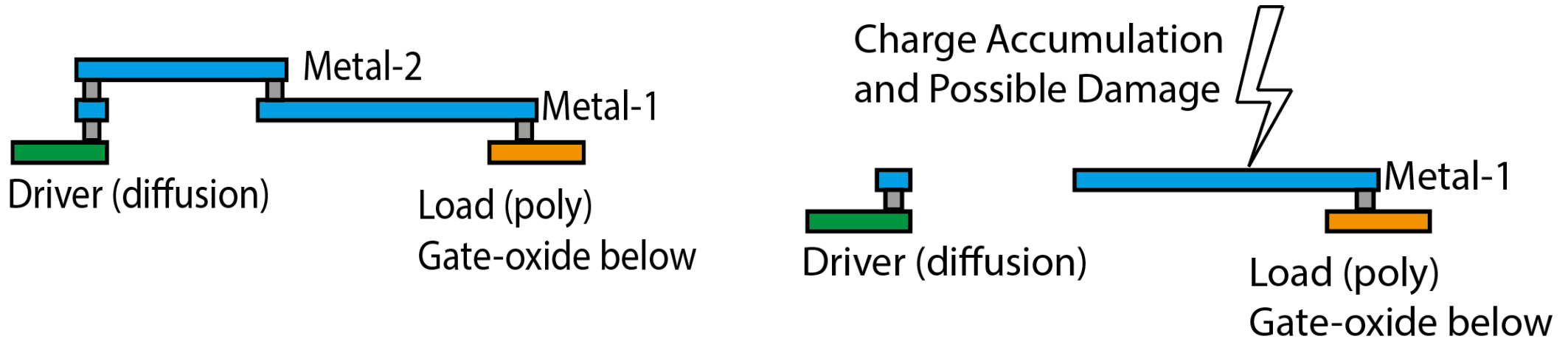
- If the magnitude of the glitch is above a threshold, it can propagate to a sequential circuit element.
 - Consequently, a wrong value can get latched (0 instead of 1 or 1 instead of 0), leading to a circuit failure.
- If the magnitude of the glitch is above a threshold, on clock/reset/control signal, a spurious transition can get triggered in a sequential element.
- Need to propagate glitches and analyze their effect in physical design

Basic Concepts for Physical Design



Antenna Effect

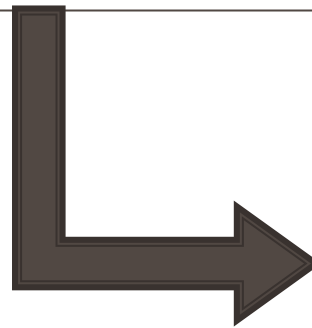
Antenna Effect: Cause



- During fabrication before addition of Metal-2, long Metal-1 connected to the gate can become charged and can destroy the gate

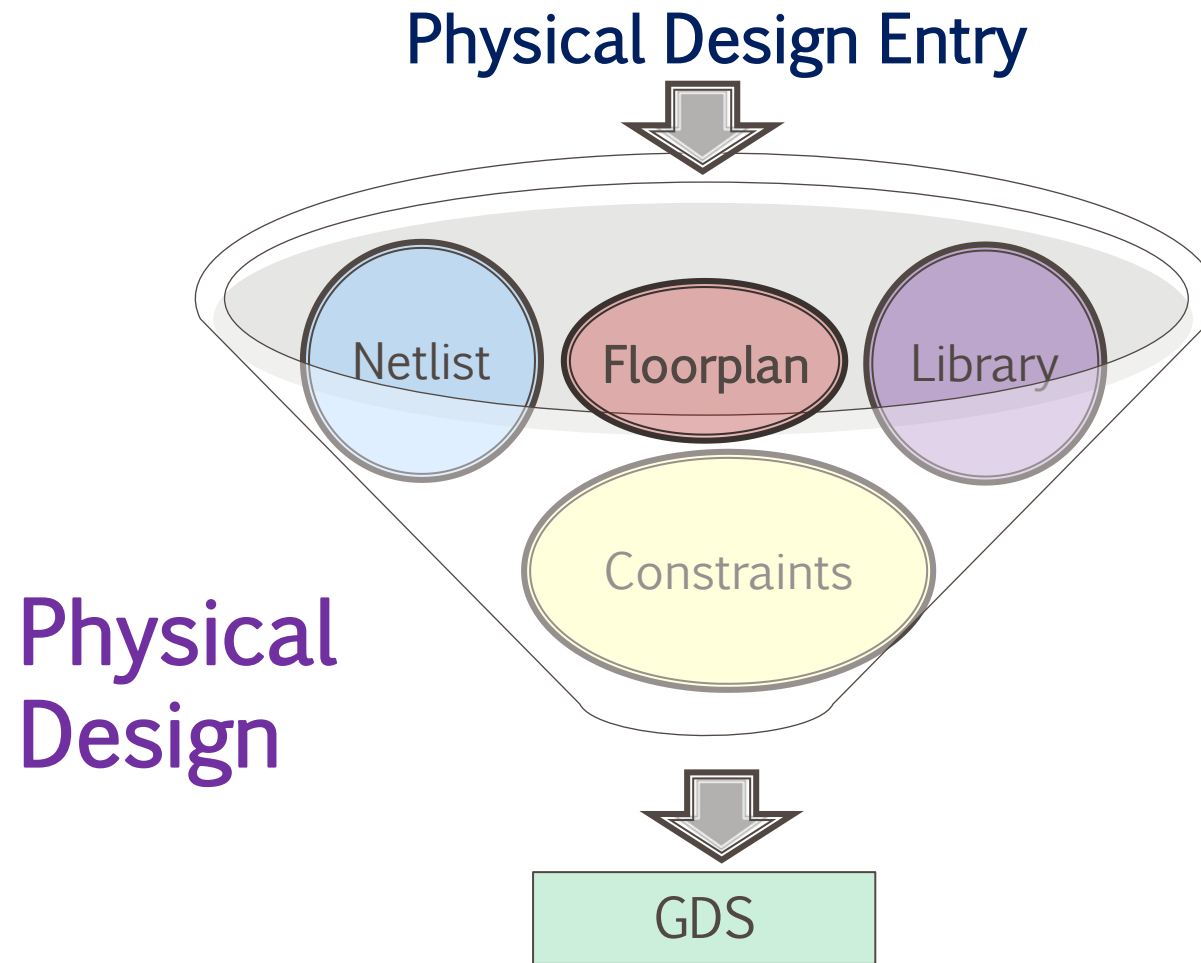
- **Antenna Ratio:** ratio of the conductor area to the gate oxide area
- High Antenna Ratio implies greater chance of damage
 - Antenna Rules defined to allow Antenna Ratio to be within limits
 - Design Goal: Fix Antenna DRC violations by keeping Antenna Ratio small

Basic Concepts for Physical Design



Library Exchange
Format (LEF)

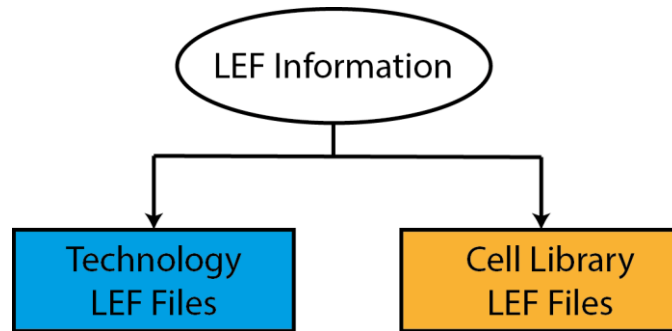
Physical Design: Inputs and Outputs



- Netlist
 - Verilog (Output of Logic Synthesis) : .v files
- Floorplan
 - Tool specific methods
- Library
 - Synopsys **Liberty** Format: .lib file
 - Abstract view of cells: *Library Exchange Format (LEF)*: .lef file (For Cadence tools)
- Constraints
 - Synopsys Design Constraints (SDC): .sdc file

Library Exchange Format (2)

- LEF file generally divided into two files



Technology LEF files:

- Information about the available layers and vias, their properties.
- Sheet resistance and capacitance per unit square for various layers
- Placement and routing design rules
- Antenna rules.

Cell Library LEF files:

- Abstract information of the cell layout relevant to the physical design,
 - Cell boundary, list of pins and their locations, and obstructions to placement and routing.

References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.